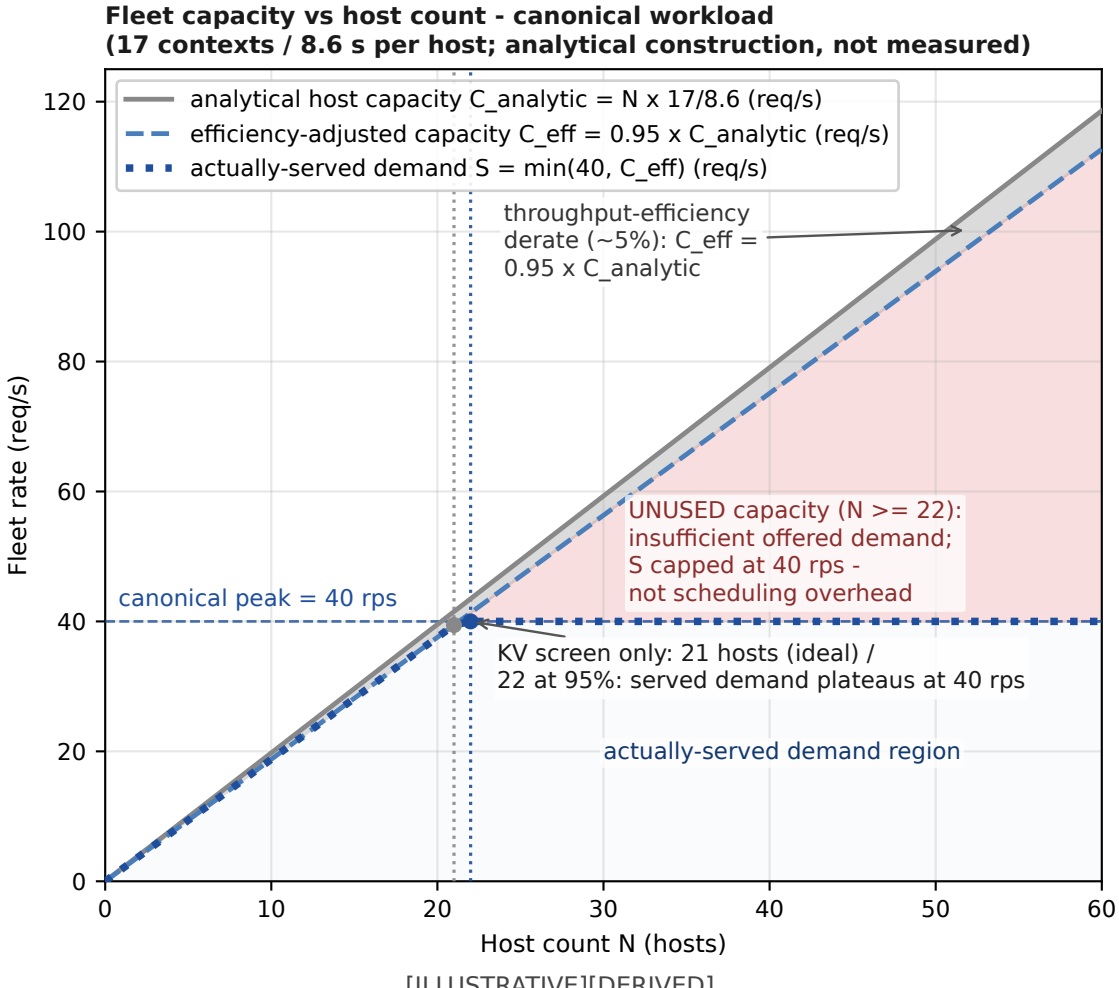
Fleet capacity vs host count - canonical workload
(17 contexts / 8.6 s per host; analytical construction, not measured)
analytical host capacity C_analytic = N x 17/8.6 (req/s)
efficiency-adjusted capacity C_eff = 0.95 x C_analytic (req/s)
actually-served demand S = min(40, C_eff) (req/s)
throughput-efficiency
derate (~5%): C_eff =
0.95 x C_analytic
UNUSED capacity (N >= 22):
insufficient offered demand;
S capped at 40 rps -
not scheduling overhead
canonical peak = 40 rps
KV screen only: 21 hosts (ideal) /
22 at 95%: served demand plateaus at 40 rps
actually-served demand region
Fleet rate (req/s)
Host count N (hosts)
0
20
40
60
80
100
120
0
10
20
30
40
50
60
[ILLUSTRATIVE][DERIVED]